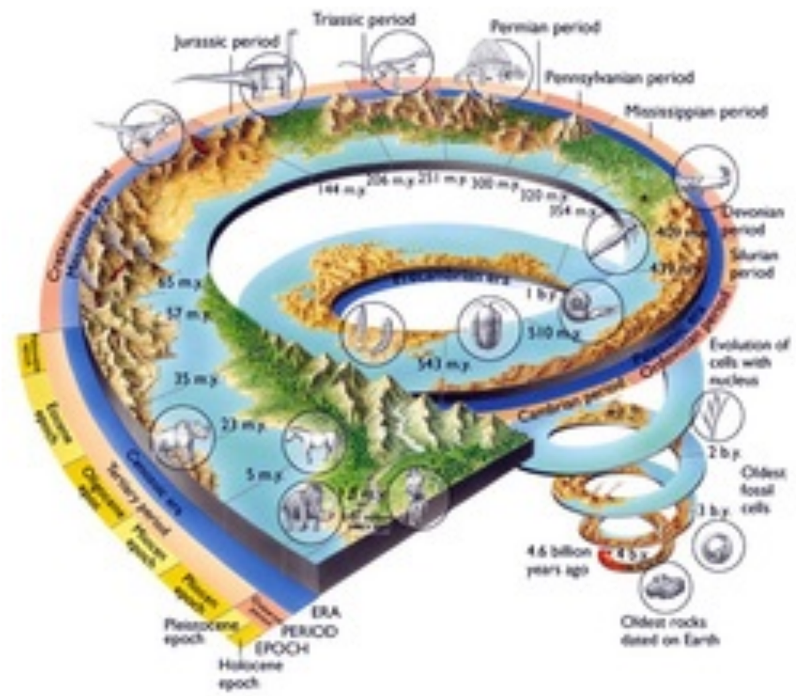
Geo2: Earth & Life story + Plate tectonics

[(https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-2-big-bang-theory)](https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-2-big-bang-theory)



Part 1: Earth's history & origin of life

When you have completed studying Part 1 of this eBook, you should be able to answer these questions:

1. How old is the Earth?
2. What were the main planetary features of early Earth at the time of the origin of life?
3. Can you graph exponential decay or growth (e.g., radioactive decay and age dating).

Part 2: Plate tectonics

When you have completed studying Part 2 of this eBook, you should be able to answer these questions:

1. What are the three types of plate tectonic boundaries?
2. What are the relative plate motions associated with each?
3. Can you draw a cross-section at a plate boundary, and locate and describe the geologic processes that occur at the boundary?

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- **The Archean (on geologic time)** (http://paleobiology.si.edu/geotime/main/htmlversion/archean1.html)
- American Natural History Museum. **"Earth Without Oxygen."** (http://www.amnh.org/explore/science-bulletins/earth/documentaries/the-rise-of-oxygen/article-earth-without-oxygen)
- US Geological Survey Publication. ***This Dynamic Earth: the Story of Plate Tectonics*** (http://pubs.usgs.gov/gip/dynamic/dynamic.html) (1996).
- NOAA Ocean Explorer: **Multimedia Discovery Missions, Lessons, 1, 2, 4** (http://oceanexplorer.noaa.gov/edu/learning/)
- Geological Society of London. **"Plate Tectonics"** (http://www.geolsoc.org.uk/Plate-Tectonics/)
- UCM Library. **"Critical Evaluation of Resources."** (http://library.ucmerced.edu/research/critical-evaluation-of-resources)

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G2P2.1

The Age of the Earth

Consider this observation:

Radiometric age dating of the oldest rocks from the Moon and the oldest meteorites found on Earth indicate an age of formation of 4.47 billion years (b.y.) ago.

So far, the oldest age of a mineral found on Earth (not in a meteorite) is 4.40 b.y., and the oldest rocks on Earth are about 4.0 b.y. old.

Why are the oldest Moon rocks and meteorites older than Earth minerals (by about 70 million years) and Earth rocks (by about 470 million years)?

What is the age of the Earth, and how is it determined?

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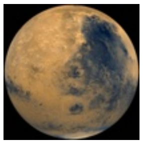
G2P2.2

The Age of the Earth

Recall that, according to the Nebular Hypothesis, all of the planets, moons, and solid bodies in the solar system formed at about the same time and from the same original starting material. Melting and differentiation of the Earth's interior occurred as a result of the Moon-forming impact, and subsequent plate tectonic processes re-surfaced and recycled the crust of the Earth back into its interior. As far as we know, plate tectonics does not occur on the Moon or other planets. Therefore, we draw two conclusions from our observations:

Earth

1. The age of formation of the Earth must be older than the Moon-forming impact, and similar to the age of formation of the other planets and solid bodies in the solar system.



Mars

2. Plate tectonics destroys and recycles the surface rocks on Earth such that most Earth rocks are young, and no rocks are present at the surface from the time of the Earth's formation.

The Moon

Based on multiple lines of evidence for the timing of the formation of the solar system, Earth, and Moon, the current best estimate of the age of the Earth is ~4.56 billion years ago.

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Hello. In this video
we're going to review the basics of radiometric age-dating and
see how it's used to determine the age of rocks and minerals over geologic time.
In order to understand this,
an important bit of chemistry to learn are the definitions of an atom,
an isotope of an atom, and what actually defines radioactivity.
Elements refer to the atoms of the periodic table,
such as carbon, oxygen, iron, etc.
Elements are the basic building blocks of all matter and
can't be broken down into simpler chemical substances.
Each element on the periodic table has an atomic number.
This number is equal to the number of protons in the nucleus and
gives each element its unique identity.
The atomic mass of an element, how much it weighs, is the sum of the protons and
neutrons in the nucleus given in atomic mass units.
So, the structure of an atom gives us the definition of an isotope of an element,
which is the same element, by def, definition having the same number of
protons, but with different numbers of neutrons in the nucleus,
and therefore slightly different atomic masses.
So, for some elements, their nucleus is naturally unstable and
spontaneously disintegrates.
This is what we call radioactivity or
radioactive decay, which changes the number of protons or
neutrons in the nucleus, and therefore changes the identity of the atom.
And it releases energy in the process.
So, let's see how this works for
radiometric age-dating and determining geologic time.
>> During the 19th century, geologists worked around the globe to deduce
the relative ages of rock layers and the fossil changes indicated over time.
By 1900, the order of events in geologic history had been fairly well-established.
But no precise manner of determining the actual span of time had yet been found.
In 1905, physicist Ernest Rutherford suggested that radioactive decay,
first discovered only nine years prior, could be used to determine the absolute or
numerical ages of rocks containing measurable of radioactive isotopes.
This is possible because radioactive isotopes, that is,
isotopes whose nuclei spontaneously eject particles and
convert to those of other elements, break down at constant rates.
Using the concept of the half-life, the consistent time it takes for
half of the atoms of a radioactive parent isotope to
convert to stable atoms of its daughter isotope, absolute ages can be determined.
If half of the parent atoms remain, one half-life has passed.
If a quarter remain, two half-lives have passed, and so on.
So, if the half-life of an isotope were 100 years, and
an eighth of the original atoms remained, three half-lives would have passed and

the rock would be 300 years old.
>> [SOUND] Okay.
You can't play with the interactive features of this graph, but I can.
Notice that this is plot of the fraction of parent atoms that are remaining in
the rock or mineral at any given time and the number of half-lives that have passed.
So, if we rewind this curve back to the beginning, no time has passed,
and we have 100% of the atoms that we started with.
As each half-life goes by, we lose half of the atoms.
So, after one half-life, we have 50% of the atoms left.
After two half-lives, we only have 25% left, and so on.
This curve describes the shape of a curve for exponential decay.
Notice that in order to determine the age of the mineral or
rock in years, you need to multiply the number of half-lives that
have passed by the years per half-life for a given isotope.
>> Examples of isotopes used to date rocks include Uranium-238 and
235, Potassium-40, Rubidium-87, and Carbon-14.
The specific isotope chosen depends on the minerals present in the rocks to be dated.
Once geologists learn to determine rock ages using the isotopic dating techniques,
they could apply these to use absolute or
numerical dates to the Geologic Time Scale.
Over the past century, the Time Scale has been refined and improved in this manner.
Hence, geologists have determined that the Earth was formed about 4.56 billion
years ago, the Phanerozoic Eon began around 542 million years ago,
and the mass extinction of the dinosaurs and
numerous other species occurred about 65 million years ago.
>> Here's the chart from the animation that you just saw,
and there's a couple of important things to note.
Different radioactive elements are found in different types of rocks and
minerals, and the radioactive parent atoms have very different half-lives.
The length of time of the half-life determines the effective age range for
dating, shown here, because you need to have enough of both the parent and
the daughter atoms left in the rock to measure.
If all of the parent atoms decay away and your radioactive clock has run out,
you have no way of knowing the amount of time that has passed.
While different minerals can trap different elements,
a favorite of geochemists is zircon, a zirconium silicate mineral
that's commonly found in a variety of different igneous and metamorphic rocks.
Zircon is a very resistant mineral and it often contains small amounts of uranium,
which decay over very long periods of time.
And so, it's useful for dating very old rocks.
Note that carbon-14 is special.
It's an isotope that can be used only to date things that were once alive, and
it has a very short half life.
So, we're limited in the age range over which we can date things with carbon-14.

I hope this introduction helps you to understand the basics behind
age-dating rocks and minerals.
Be sure to review the chemical definitions of atoms, isotopes and radioactivity.
You'll also have an exercise in your lab section that'll give you
some practice in working with these concepts.
English
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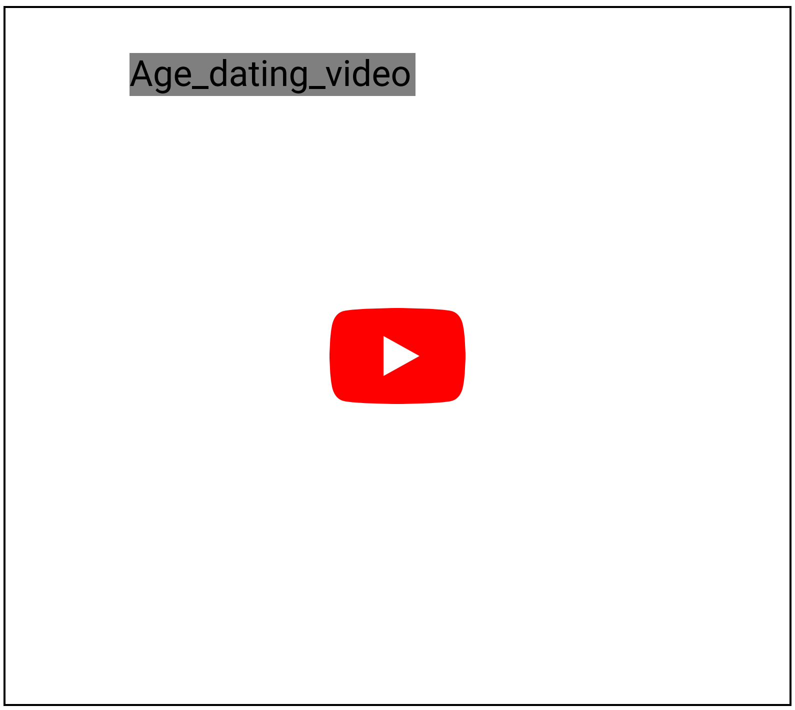
G2P2.3

How Do We Determine the Age of Rocks and Minerals?



Minerals contain very small amounts of naturally occurring radioactive isotopes of elements, some of which are very long-lived and take billions of years to decay. When these radioactive isotopes are trapped in minerals as they crystallize, they act as atomic time-keepers of the age of formation.

View this animation [7:00] about radioactivity and radiometric age dating and read the information on the web site below.

Be sure you understand the concept of a "half-life" of a radioactive isotope and how this allows us to determine the age of a mineral or rock.

Also watch the video on the following website:

Smithsonian, **How Do We Know the Earth Is 4.6 Billion Years Old? (https://www.smithsonianmag.com/smart-news/how-do-we-know-earth-46-billion-years-old-180951483/)**

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ASK SMITHSONIAN 2017

A *Smithsonian* magazine special report

SMART NEWS

Cool Finds

How Do We Know the Earth Is 4.6 Billion Years Old?

We know the Earth is old. But how do we know its age?

Colin Schultz

May 16, 2014

How To Date A Planet

Watch on

The process of figuring out a rock's age often falls to the scientific techniques of radiometric dating, the most famous of which is radiocarbon dating. This process focuses on the ratio between the number of carbon-14 and carbon-12 isotopes in any once-living being: that ratio indicates how long it's been since that being was alive. But carbon is not the only element that can be dated—a whole host of others exist. In uranium-lead dating, for instance, the radioactive decay of uranium into lead proceeds at a reliable rate.

Based on the very old zircon rock from Australia we know that the Earth is at least 4.374 billion years old. But it could certainly be older. Scientists tend to agree that our little planet is around 4.54 billion years old—give or take a few hundred million.

Colin Schultz 𝕏

Colin Schultz is a freelance science writer and editor based in Toronto, Canada. He blogs for Smart News and contributes to the American Geophysical Union. He has a B.Sc. in physical science and philosophy, and a M.A. in journalism.

Filed Under: Cool Finds, Earth Science

G2P2.4

Review: Radiometric Age Dating

Radioactive isotopes:

- Are found naturally in small amounts in minerals and rocks.
- Carbon-14: Can be used to date only organic matter from something that was once alive, and only for relatively recent ages (< 70,000 years).
- An isotope of appropriate half-life must be used to date rocks, fossils, or organic material – the rock or fossil must have enough of the parent and daughter isotopes left to measure!

This means that:

- You had to have enough of the parent isotope to start with.
- You did not lose any parent or daughter isotopes over time.
- The rock cannot be too old for the isotope clock, or all of the parent atoms will have decayed away.

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G2P2.5

What is the age of the oldest preserved rocks on Earth?

Thus far, the oldest rocks , found near Hudson Bay in the Canadian arctic, are about 4.0 billion years old, younger than the oldest Moon rocks and meteorites.

Zircons found in rocks from the Jack Hills in western Australia have been dated at 4.40 billion years old using U-Pb isotopes (the rest of the rock is younger).

Figure 8.15a
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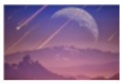
G2P2.6

Conditions on Early Earth

What was Earth like during this early time when life emerged?



Although we don't know exactly what the Earth was like during its first billion years (the “Hadean” and early “Archean” Eons), we have some clues from the study of other planets, computational models, and from the oldest rocks found so far on Earth. Of course, one of the biggest differences was that the only life on Earth for its first several billion years was single-celled organisms.

Read

The following web articles about these geological periods:

1. **The Hadean (~4.6 – 4.0 b.y.) (http://palaeos.com/hadean/index.html)**
2. **The Archean (~4.0 – 2.5 b.y.) (http://palaeos.com/archean/index.html)**

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Palaeos *Life Through Deep Time*	Παλαιός	The Hadean Eon
TIME		THE HADEAN EON

Page Back:	Unit Home: (You are here)	**Included in: Precambrian**	**Main topic: Timescale**	**Page Next: Hadean**
Timescale Back: Chaotian (earlier)	**Planetary timescales** (parallel)	**Includes: Hadean overview** (more detail)	**Earth** (formed during Hadean)	**Timescale Next: Archean** (later)

The Hadean Eon

The Hadean Eon of Precambrian Time: 4520 to 3800 million years ago

Timescale **Chaotian** **Hadean** **Palaeohadean** **Mesohadean** **Neohadean** **Archean** **Proterozoic** **Phanerozoic** Google Search only search Palaeos.com	**The Hadean Eon** **The Hadean Eon** **Overview of the Hadean** **Hadean Timescale** **Notes and References**

Surface of the Earth during the Hadean eon, presumably the Hephaestean period) - Copyright 1994-2011, Streamlined Technology, LLC and Barewalls Interactive Arts, Stocktrek Images original url
see also artwork by Hermann Trappman

The Chaotian Eon ended with a collision between the young proto-Earth and an almost Mars-sized body (which has been named Theia after the mother of the Moon goddess Selene) (Halliday, 2000) . This ejecta from this gigantic collision eventually formed into the moon, whilst denser matter from both protoplanets ended up as the Earth. The originally molten Earth soon cooled, although the Earth's surface was continually bombarded by meteorites, and the much hotter mantle would have resulted in severe volcanism. At this time, the Earth would have perhaps looked rather like the above drawing, an alien and inhospitable world, which is the Hadean in the popular imagination. However there very soon formed oceans or seas (either through comets or volcanic outgassing), and an atmosphere, and for most of the Hadean, conditions were milder. It is possible that some form of microbial life may have also appeared. If it did it is not certain whether it would have survived the period of intense meteorite impacts known as the "Late Heavy Bombardment" (Gomes et al 2005), , although the tenacity of extromophile microbes might argue in favour. The end of this period, around 3.8 billion years ago, marked the end of the Hadean and the beginning of the Archean eon.

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Palaeos *Life Through Deep Time*	Παλαιός	*The Archean Eon*
TIME		THE ARCHEAN EON

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Unit Back: Hadean (Timeline - earlier)	Unit Back: Tellurobiota (Life - earlier)	Unit Back: Earth	Includes: Archean overview (more detail)	Unit Next: Bacteria (Life - later)	Unit Next: Proterozoic (Timeline - later)

The Archean Eon

Google Search
only search Palaeos.com

- Timescale
  - Chaotian
  - Hadean
  - Archean
    - Eoarchean
    - Paleoarchean
    - Mesoarchean
    - Neoarchean
  - Proterozoic
  - Phanerozoic

- Archean
  - The Archean Eon
  - References

Archaean Landscape; image from Space Biology - source, NASA.

The Archean (or Archaean; formerly Archeozoic or Archaeozoic) was a long and, due to its great distance in time, poorly known geologic eon that last a whopping 1.3 gigayears (or 13 geons). That's almost two and half times the entire history of complex life on Earth. Yet a lot was happening at this time; the origin and growth of continents, and the diversification of life (which may or may not have appeared as early as the preceding Hadean). For this entire time span, and for much of the following Proterozoic Eon, the highest form of life on Earth were algae mats, which sometimes formed dome like structures called stromatolites. These are shown somewhat fancifully in the illustration above, in fact the stromatolites would have actually been under several meters of water; the lack of oxygen and hence of a protective ozone layer in the Precambrian atmosphere meant that the sun's rays would be deadly to any unprotected life.

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G2P2.7

The Faint Young Sun

In the early 1970's, Carl Sagan and George Muller pointed out an interesting "paradox":

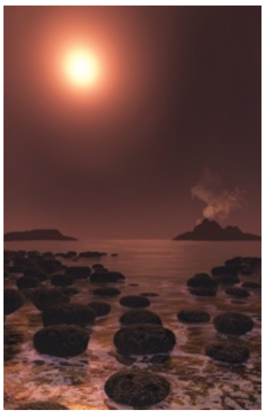

Image: *Science News*

Four billion years ago after the Earth and other planets formed, the Sun, being a younger star then, would have been about 30% less bright than it is today, and it has been growing brighter ever since. This means that 30% less solar radiation would have reached the Earth, which is enough of a reduction in solar radiation for water at the Earth's surface to freeze.

But this is the around the time that life emerged on Earth, and there is geologic evidence for sediments and microbial life around 3.8 billion years ago – evidence that liquid water was present and the planet was not frozen.

So how did the Earth stay warm enough for water to remain a liquid and for life to begin?

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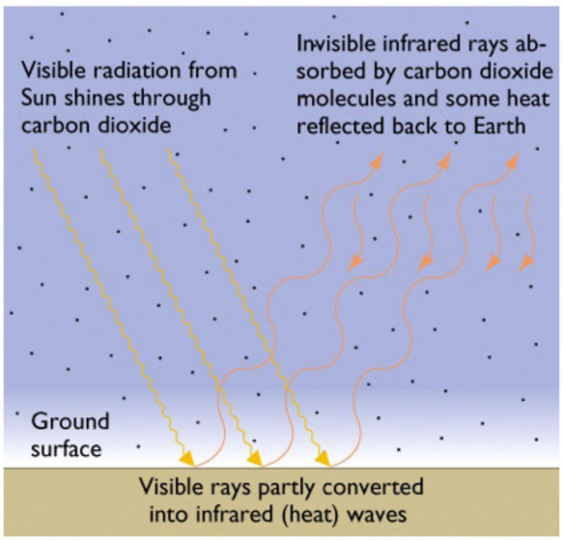
G2P2.8

Temperature on Early Earth



In order for water at the Earth's surface to not freeze during the early period of low solar luminosity, many scientists think that the Earth was warmed by a higher concentration of "greenhouse" gases at this time, similar to the warming that our planet is experiencing today from human contributions of carbon dioxide (CO_2), methane (CH_4), and other gases. Incoming solar radiation (sunlight) is only partially absorbed by the atmosphere. Radiation that hits the Earth's surface is adsorbed and radiated back to the atmosphere mostly as infrared radiation – heat. Carbon dioxide and other greenhouse gases adsorb infrared radiation more efficiently than solar radiation, and trap this heat in the atmosphere.Some models suggest that Earth's surface temperature was considerably warmer than today during the early Archean, which would require a very high concentration of atmospheric CO_2. Other models point out that if the early Earth was mostly ocean with a small amount of land mass and fewer clouds (a low albedo), the dark ocean would adsorb more heat and kept the planet warmer. Whatever the explanation, water was in liquid form and life emerged by around 3.8 billion years ago.

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G2P2.9

Earth's Early Atmosphere

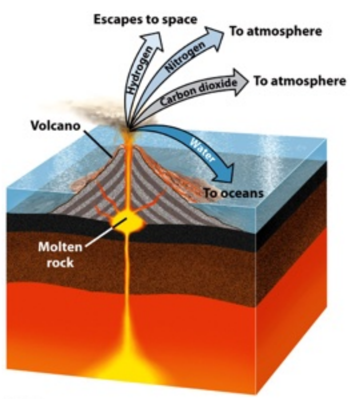

Figure 9.6
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The Moon-forming impact, and likely other very large impacts during the period of heavy bombardment, would have driven off gases in the atmosphere, perhaps multiple times. These impactors, however, might also have delivered gases such as CO_2 and H_2O, to Earth by degassing of their solid material after impact. Other sources of volatile species to the early atmosphere may have been volcanic activity, which could have added gases such as CO_2, H_2O, and CH_4 (methane) and H_2S (hydrogen sulfide), or weathering of rocks exposed on early continents.

We don't know the exact composition of the early atmosphere, but multiple lines of evidence indicate that our present-day atmosphere is of secondary origin, and that the earliest atmosphere was quite different for at least the first billion years of Earth history. While the compositional details are a matter of debate among scientists, there is agreement that there was very little or no free oxygen (O_2) in the atmosphere before about 2.7 billion years ago.

Reading

Changes in the early Earth's crust caused oxygen to fill the atmosphere (https://catcourses.ucmerced.edu/courses/32116/files/7102690?wrap=1)
(https://catcourses.ucmerced.edu/courses/32116/files/7102690/download?download_frd=1)

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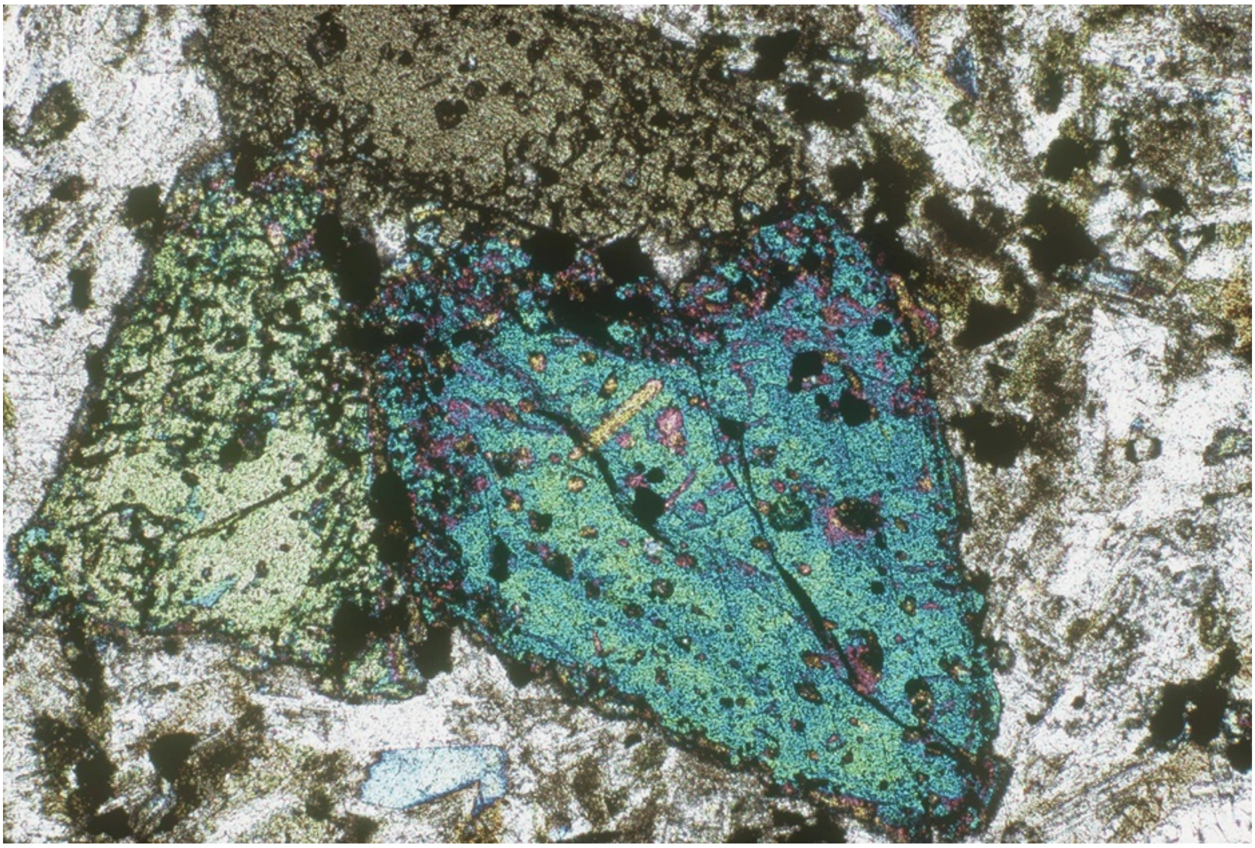
Rocks, not bacteria, triggered Earth's oxidation

Changes in the planet's crust paved the way for an oxygen atmosphere and the evolution of complex life, writes Andrew Masterson.



Olivine basalt. When the highly reactive olivine became less common in Earth's crust, the oxygen produced by cyanobacteria transformed the atmosphere.
CREDIT: RAY SIMONS / GETTY

The Great Oxidation Event – the period 2.4 billion years ago when the Earth's atmosphere suddenly became rich in oxygen, thus catalyzing the evolution of aerobic life – was caused by geological, rather than biological, change, new research has found.

Geologists Matthijs Smit from the University of British Colombia in Canada and Klaus Mezger from Switzerland's University of Bern set out to trace the cause of the oxidation event, which saw the amount of oxygen in the oceans increase 10,000 times in just 200 million years – an incredibly rapid rise.

At the time, cyanobacteria were already in existence and producing oxygen as a byproduct of photosynthesis. Some studies have suggested that this process was solely responsible for the sudden rise in available oxygen, but there has been **little data to support the contention <http://www.pnas.org/content/110/5/1791.short>** .

Smit and Mezger decided to take a different approach and looked to rock formations for a possible answer. During the Great Oxidation Event the rocky composition of the continents also underwent significant chemical change.

At the start of the event, most shales and igneous rock types around the world comprised high levels of magnesium and low levels of silica. This is a similar composition to rocks found today in volcanically active areas such as Iceland.

RECOMMENDED

Bacteria's evolution sheds light on great oxygenation event

BIOLOGY

These ancient formations, however, also contained another very plentiful mineral called olivine, a type of magnesium iron silicate. When it comes into contact with water, it catalyses chemical reactions that lock away oxygen, meaning it is not freely available as a gas – which is essential for aerobic respiration.

Indeed, the scientists calculate that olivine was plentiful enough that it would have tied up all the oxygen produced by cyanobacteria, neutralising its effect.

However, at the same time as the oxidation event was occurring, the continental crusts were evolving. Their composition changed – in particular, olivine became very rare.

As a result, fewer olivine reactions meant that oxygen produced by cyanobacteria was gradually able to accumulate in the oceans without being tied up. Eventually it saturated the water and began to leach out into the atmosphere.

"It really appears to have been the starting point for life diversification as we know it," says Smit.

"After that change, the Earth became much more habitable and suitable for the evolution of complex life, but that needed some trigger mechanism, and that's what we may have found."

The research is **published in the journal <http://dx.doi.org/10.1038/ngeo3030> *Nature Geoscience* *<http://dx.doi.org/10.1038/ngeo3030>*** .

__ANDREW MASTERSON__ is news editor of Cosmos.

G2P2.10

Major Differences Between Early Earth (~4.5 to 3.8 b.y.) and Present-day Earth

- Solar luminosity was about 25-30% less than today, and therefore less solar radiation hit the Earth's surface (Sagan's "Faint Young Sun Paradox")
- Geochemical evidence suggests that surface temperatures were likely higher on average, although there is disagreement about how much
- There was no free oxygen molecules (O_2) in the atmosphere, but probably a higher concentration of CO_2 and perhaps methane, which produced global warming from the greenhouse effect; a lower surface albedo (a darker, less reflective surface) would also help warm the planet
- Because there was no oxygen in the atmosphere, no ozone (O_3) was produced in the stratosphere, and therefore more ultraviolet radiation from the Sun would reach the Earth's surface
- There was a much higher frequency of meteorite and asteroid impacts until about 3.9 b.y., some of which were likely quite large
- Heat flow from Earth's interior is calculated to be higher, but not nearly high enough to compensate for the lower amount of solar radiation

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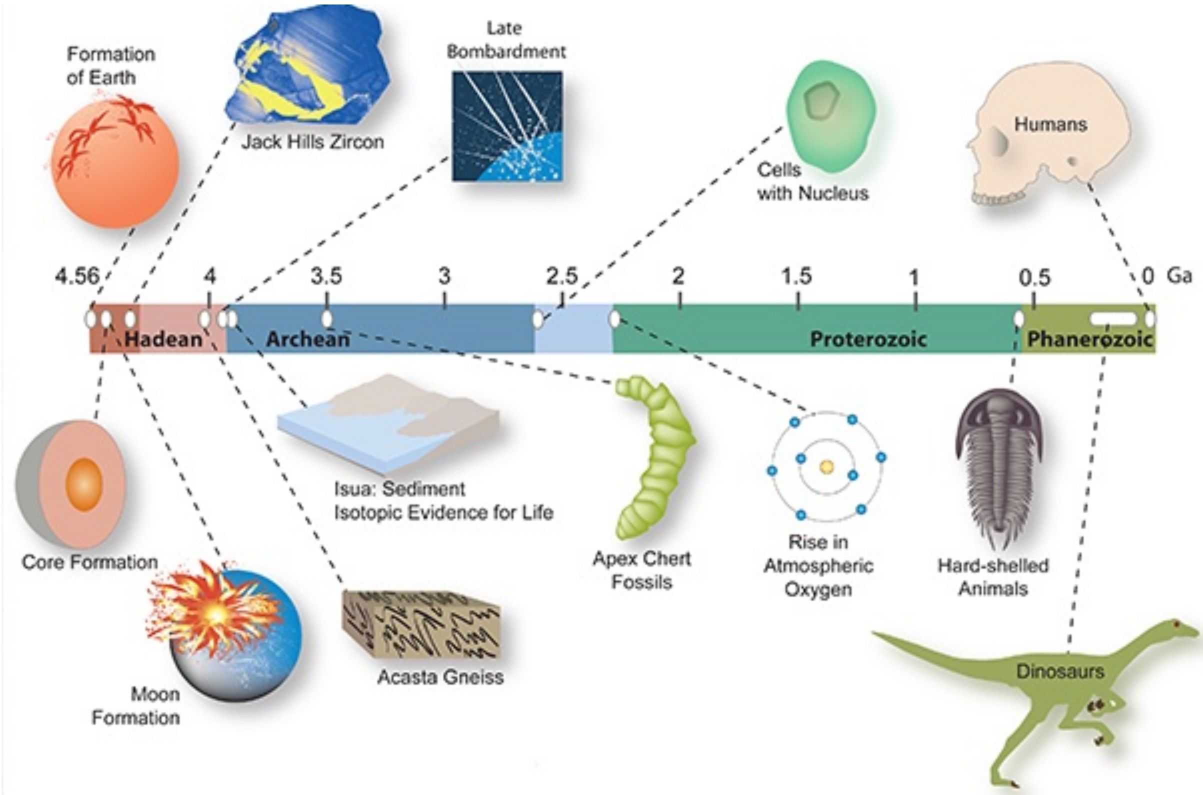
G2P2.11

Emergence of Life on Earth

The timeline of events as we currently know them...



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G2P2.12

Summary

Although many questions remain regarding the exact circumstances that enabled life on Earth to emerge, several lines of evidence indicate that it occurred relatively early in geologic time after the Moon-forming impact, likely before 3.8 b.y. when we have isotopic evidence in rocks that points to the presence of living organisms. The existence of zircon with an age of 4.4 b.y., and rocks that are at least 4.0 b.y. old, indicate that crustal processes were occurring at this time that were not much different from present day – that is, it is likely that plate tectonics began very early in Earth's history.

This concludes Part 1 of Chapter 2 of the module Geosphere. Let's start now part 2, the final section of Chapter 2.

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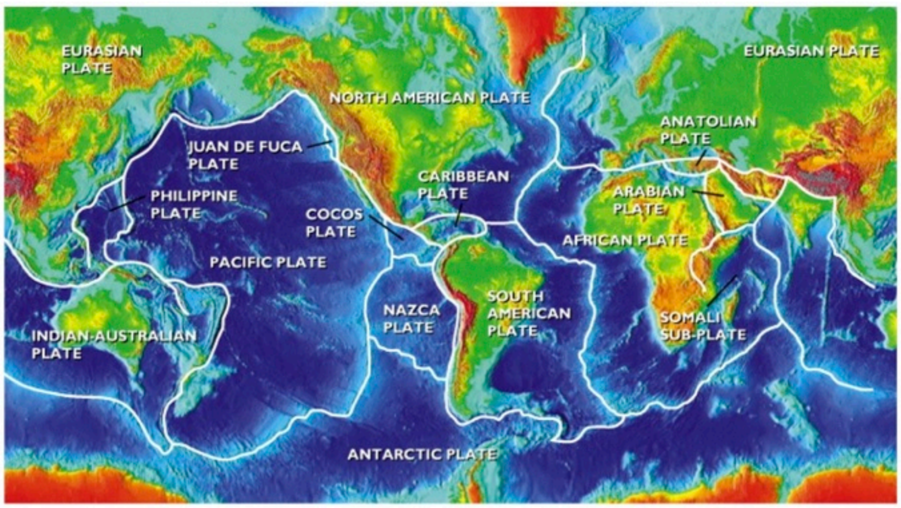
G3.1

Introduction to Plate Tectonics

This lesson presents the main principles of plate tectonics and describes the physical and geological features associated with the three types of plate boundaries. As a general scientific theory for describing the behavior of the solid Earth, plate tectonics is remarkable for its ability to explain and unify a wide range of geologic processes.



There is a wealth of information and instructional sites available online about plate tectonics and basic geology. In this lesson and the next one, we provide links to a few external sites to guide your study. In your own online exploration, you may find other sites with accurate information from which to learn this material. Feel free to explore, but be critical about online site quality – see the next page for some tips.

For more in-depth reading about plate tectonics, see the online version of this US Geological Survey Publication, ***This Dynamic Earth: The Story of Plate Tectonics*** ***(http://pubs.usgs.gov/gip/dynamic/dynamic.html)*** (published in 1996, but still accurate, relevant, and well-written).

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G3.2

Sources of Information and Attribution

How do you know if an online source of information is reliable?

1. ***Consider the owner/author of the site as you weigh the credibility, objectivity and value of the information:***
   - Is the website owner an established organization?
   - A government agency?
   - An educational institution? An impartial agency?
   - An advocacy group?
   - A commercial company?
   - An individual?

2. ***Consider "peer-review:"*** **Have the articles or information been reviewed by impartial peers:**
   - Is it a scholarly journal article?
   - An academic source?
   - A reputable newspaper?

3. ***Assess the Quality of the information:***
   - Are sources of data well documented?
   - Is the information current?
   - Is there evidence of bias?

4. ***Seek multiple sources:***
   - Don't rely on one source of information for a topic.
   - Consider whether multiple, reliable sources agree when using fact-based arguments.
   - Do they agree on facts and data even when interpretations differ?

When completing written assignments remember to cite your sources of information. Whenever possible, use the primary source of the statement or fact, not a secondary compilation such as wikipedia. Be critical! For more guidance, see the UCM library page, **Critical Evaluation of Sources (http://library.ucmerced.edu/research/critical-evaluation-of-resources)** .

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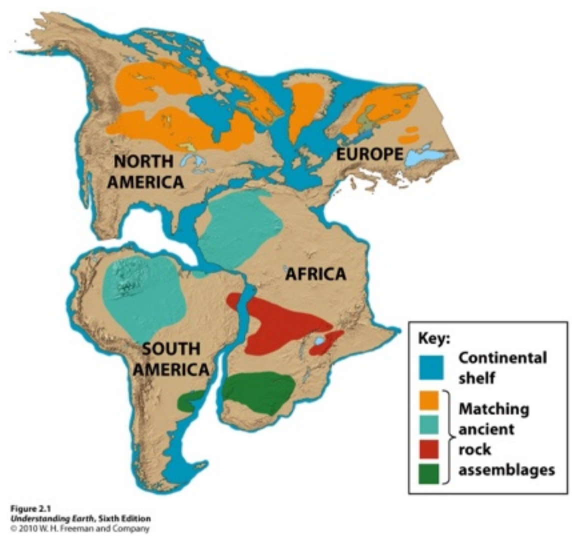
G3.3

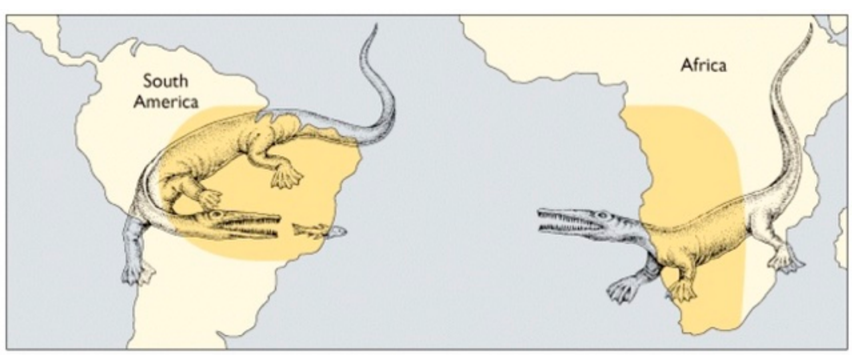
Wegener’s Hypothesis: The Idea Of “Continental Drift”



In 1915, Alfred Wegener noticed that the continental margins of South America and Africa, and North America and Europe, could be fit back together like pieces of a puzzle. Geologic studies and paleontological evidence revealed that similar rock assemblages and characteristic, extinct fossil plants and animals matched between the continents now separated by the Atlantic Ocean.

He proposed that the continents were once joined together and then drifted apart over geologic time (referred to as “Continental Drift”). Wegener’s hypothesis was not widely accepted in the early 20th century because there was no plausible explanation for how continents could move across vast ocean basins.

What was not known at the time, because no one had explored the ocean basins, was that the entire crust of the Earth was broken into pieces or “plates”. The continents were not moving through the oceans, but rather, the entire crust of the Earth was shifting around over long periods of geologic time.



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G3.4

Solid Earth: Earth’s Major Layers

Earth_layers

The inside of the Earth is layered or differentiated, both chemically and physically. This internal structure is intimately linked to plate tectonics, both as a result of billions of years of tectonic processes, and as the driving force underlying plate tectonics.

Earth’s interior can be divided into three distinct chemical or compositional layers, which gives rise to different physical or mechanical behavior:

1. Crust:
   Oceanic 0-6 km (“young”, < 180 m.y.)
   Continental 0-34 km (older, up to ~4.0 b.y.)
2. Mantle:
   Upper 34-670 km
   Lower 670-2900 km
3. Core:
   Outer (liquid) 2900-5160 km
   Inner (solid) 5160-6370 km

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The Earth is layered.
The layering within our planet is both chemical, that is,
layers of different rock compositions, at different depths, and mechanical.
There are at least five distinct layers in the Earth
that have significantly different physical properties,
with respect to whether they break, bend, or flow under stress.
Chemically, the Earth can be divided into the crust, mantle, and core.
The density of these three layers increases as you move inward,
with the crust being the least dense, and the core being the most dense.
The Earth's continental crust, is made primarily of rocks similar to granite.
A light-colored crystalline rock, rich in aluminum, sodium, potassium, and silicon.
While the oceanic crust is primarily made of basalt and gabbro,
dark-colored igneous rocks rich in iron, magnesium, calcium, and silicon.
The mantle is made primarily of crystalline rocks.
Very rich in iron, magnesium, and silicon.
The core is metallic and is made primarily of iron, and
to a much lesser extent, nickel.
In addition to the three compositional layers previously mentioned,
the Earth can also be divided into five mechanical layers,
based on differing responses to forces or stresses.
The lithosphere is the relatively cold, brittle outer shell of the Earth,
composed of the entire crust, and a bit of the uppermost mantle.
The lithosphere tends to crack or fracture, under stress.
The asthenosphere is within the uppermost mantle, directly beneath a lithosphere,
and is gooey and partially molten, allowing it to flow slowly, under stress.
The deep mantle, also known as the mesosphere, is a very hot plastic solid,
that can flow extremely slowly over many millions of years.
Beneath the mesosphere is the outer core, composed of liquid metal,
which flows easily.
In the very center, is the inner core, which is a white hot sphere of metal,
kept solid only because of the extreme pressure it is under
due to the weight of all the other layers above it.

G3.5

Density Differences: Continental and Oceanic Crust

Lithosphere (= crust + upper mantle) is strong, brittle, and less dense. It floats on hot, weaker, and more plastic asthenosphere (= upper mantle) that is denser but partially melted.

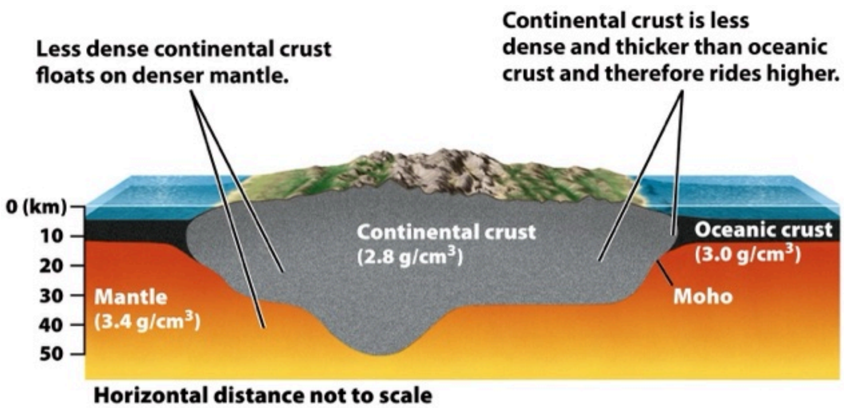

Figure 1.11
Understanding Earth, Sixth Edition
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Practice Problem	
$\text{Density} = \dfrac{\text{Mass of an object}}{\text{Volume of an Object}}$ $\text{Units} = \dfrac{\text{Mass (g)}}{\text{Volume (cm}^3\text{)}}$	*The crust of the Earth – the thin outer shell – makes up only 0.4% of Earth's total mass. The total mass of the Earth is 6 x 10^{27} g. If the average density of the crust is 3 g/cm^3, what is the total volume of the crust?*

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G3.6

Practice Problem Solution

$$\frac{\text{Density (g)}}{(\text{cm}^3)} = \frac{\text{Mass (g)}}{\text{Volume (cm}^3)}$$

The crust of the Earth – the thin outer shell – makes up only 0.4% of Earth's total mass. The total mass of the Earth is 6×10^{27} g. If the average density of the crust is 3 g/cm^3, what is the total volume of the crust?

$$\text{Volume (crust)} = \frac{\text{Mass (g)}}{\text{Density (g/cm}^3)} = \frac{6 \times 10^{27} \text{ (g)}}{3 \text{ (g/cm}^3)}$$

$$\text{Volume (crust)} = 2.0 \times 10^{27} \text{ cm}^3 \times 0.004$$
$$= 8.0 \times 10^{24} \text{ cm}^3$$

Note: to multiply by 0.4%, convert to 0.004

6

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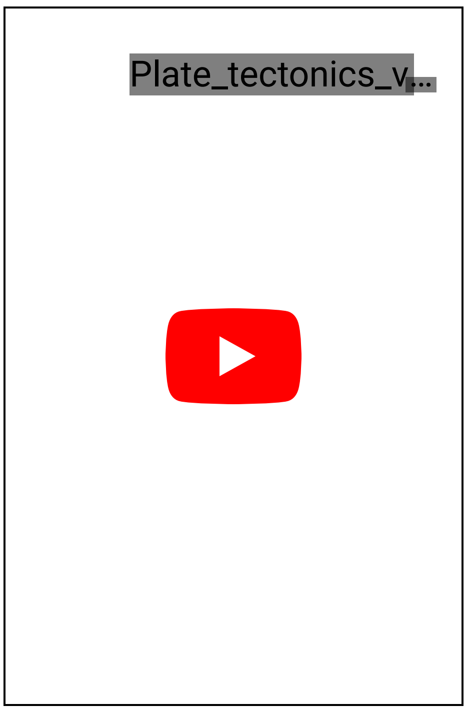
G3.7

Plate Tectonics: Plates and Their Boundaries

Plate tectonics was proposed in the 1960's as a general, unifying theory of the solid Earth. It was a revolutionary advancement in Earth sciences and served to link and explain many diverse geologic and geophysical observations. In general, plate tectonics:

- Assumes a stiff lithosphere (crust) floating on top of weak asthenosphere (mantle) Explains the motion and behavior of Earth's lithosphere
- Explains mechanisms of continental drift, seafloor spreading, and subduction
- Can be used to rationalize and predict the locations of earthquakes and volcanoes



View the video [3:30] introduction to plate tectonics. Then, view **"Plate Tectonics" (http://oceanexplorer.noaa.gov/edu/learning/player/lesson01.html)** from NOAA, which covers the same information in greater scientific detail.

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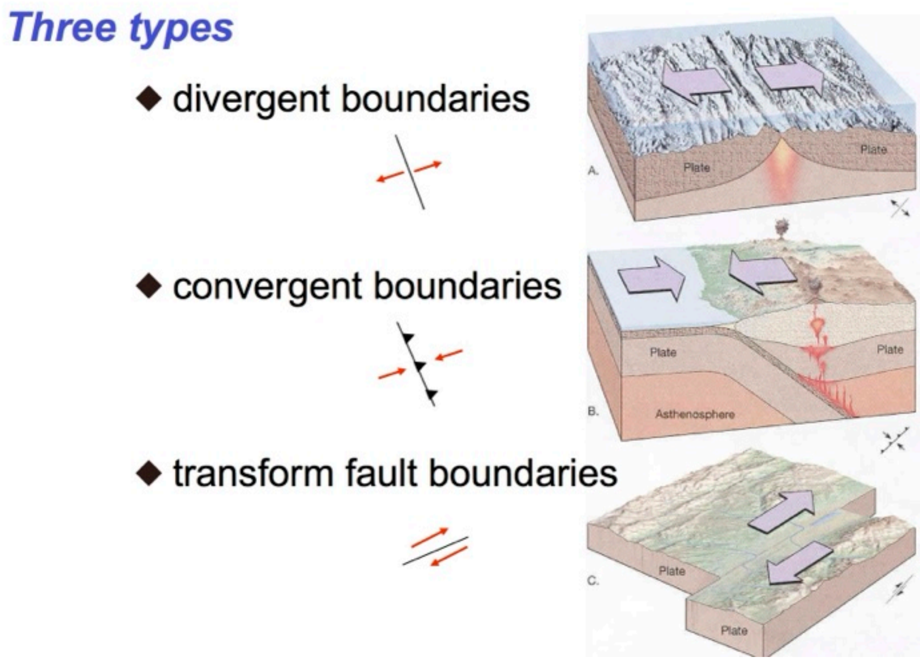
Like thin ice on a pond, the brittle lithosphere
is broken into segments called "plates."
The modern lithosphere is fractured into seven major plates and
several minor ones.
The plates move by floating horizontally over the soft plastic asthenosphere.
Today these plates are moving at rates between
1 and 20 centimeters per year, in the directions shown by the arrows.
Where any two plates meet, their motions cause them to bump and
grind together at their boundaries.
As a result, most earthquakes, volcanoes and
mountain ranges concentrate at the plate boundaries.
As the plates move horizontally, floating over the asthenosphere,
they move in different relative directions where two plates meet.
In some places, two plates spread apart from one another.
This kind of plate boundary, is call a spreading center.
At a spreading center, new lithosphere forms as hot asthenosphere rock rises
to fill the gap left by the separating plates.
Some of the hot asthenosphere melts as it rises, forming basalt magma.
Most of this magma rises to the surface to form new oceanic crust.
The new lithosphere at the spreading center is hot and
therefore of low density.
Because of its low density it floats to higher level than
the surrounding sea floor, forming a mid-oceanic ridge.
As the oceanic crust separates, blocks of rock slip downward,
forming a rift valley in the center of the mid-oceanic ridge.
Beneath the seas, the mid-oceanic ridge circles the Earth for 80,000 kilometers.
It's the longest, largest mountain chain on Earth.
As new lithosphere spreads outward from a spreading center, equal amounts of
old lithosphere disappear by sinking into the mantle at a subduction zone.
Here, where two plates collide, one dives into the mantle beneath the other.
The diving plate drags part of the sea floor down with it
forming an oceanic trench.
Trenches are the deepest parts of the seas,
reaching depths of more than ten kilometers.
Great quantities of basaltic magma form in a subduction zone.
The magma rises to erupt onto the sea floor.
It forms volcanoes that eventually rise above sea level, to create an island arc.
Frequent earthquakes occur in the subduction zone
where the diving plate scrapes and jerks its way down into the mantle.
Island arcs formed in this way are abundant in the Southwestern Pacific
Ocean, where subduction is occurring today and volcanic eruptions are common.
Subduction can also bring a diving plate beneath the edge of a continent.
Basaltic magma forms here, too.
But something different happens when it rises into the base of the continent.

Continents are made of granite.
The salt magma is hundreds of degrees hotter than the temperature at
which granite melts.
Therefore when the basalt magma rises into the base of the continent,
it melts the granite, forming huge amounts of granite magma.
The granite magma then rises and solidifies higher in the crust
to form great granite plutons.
The addition of so much magma and heat, causes the crust to swell and thicken,
forming a great mountain chain above the subduction zone.
The mountains erode as they rise.
Eventually the granite plutons are exposed at the Earth's surface.

G3.8

Plate Tectonics: Plates and Their Boundaries



We will now look in more detail at each of the three types of plate boundaries. Keep in mind the physical and geologic features that distinguish each boundary: mountains and trenches, locations and types of earthquakes and volcanoes, and plate motions.

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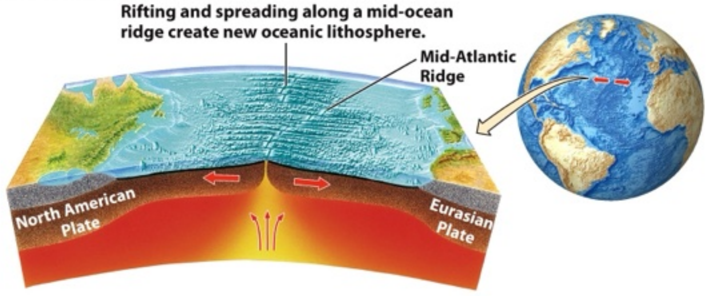
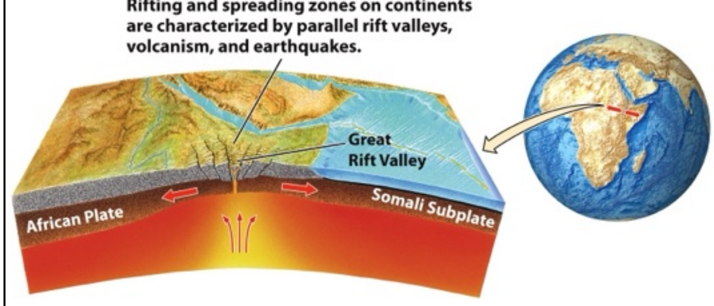
G3.9

Divergent Boundaries

There are two general types of divergent boundaries:

Oceanic Spreading Center	Continental Rift Zone
 Figure 2.8a Understanding Earth, Sixth Edition © 2010 W. H. Freeman and Company	 Figure 2.8b Understanding Earth, Sixth Edition © 2010 W. H. Freeman and Company

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View

NOAA Ocean Explorer: Mid-ocean Ridges (Lesson 2) (http://oceanexplorer.noaa.gov/edu/learning/player/lesson02.html)

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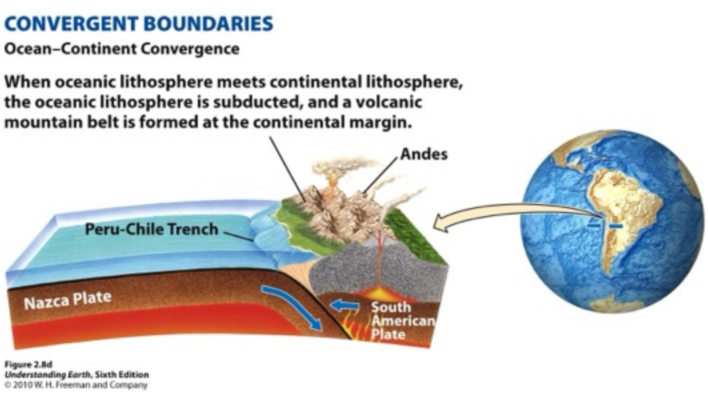
G3.10

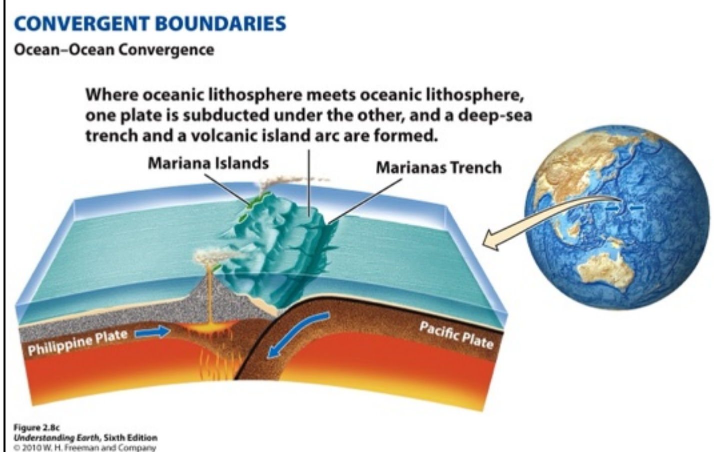
Convergent Boundaries

Go to the NOAA site and watch this video convergent plate boundaries:
[**NOAA Ocean Explorer: Subduction Zones (Lesson 4)**](http://oceanexplorer.noaa.gov/edu/learning/player/lesson04.html) **(http://oceanexplorer.noaa.gov/edu/learning/player/lesson04.html)**

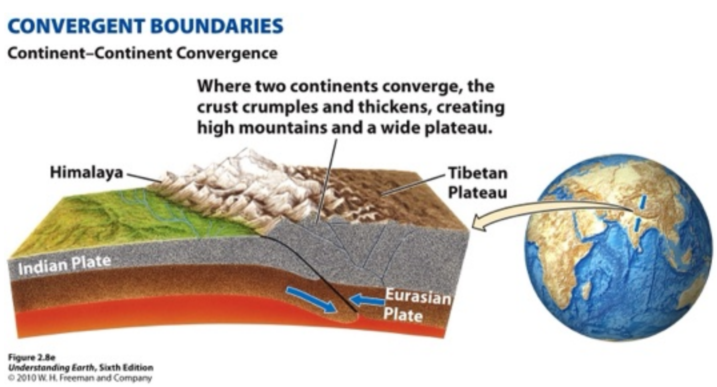
Ocean-Continent Collision



Ocean-Island Arc Collision



Continent-Continent Collision



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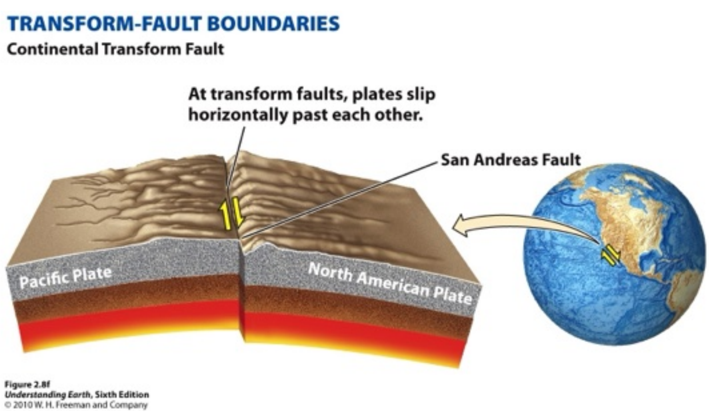
G3.11

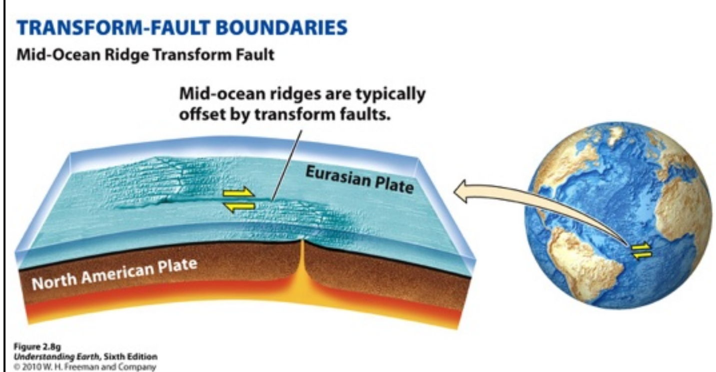
Transform Fault Boundaries

There are two general types of transform fault boundaries:

Continental



Mid-Ocean Ridge



View

"Transform Boundaries" (http://youtu.be/A7BkG5KoLGU) [2:00] features animations that summarize the features of transform boundaries. For additional information on transform boundaries consult the **Geological Society of London's webpages on the San Andreas fault and plate tectonics. (http://www.geolsoc.org.uk/Plate-Tectonics/Chap3-Plate-Margins/Conservative/San-Andreas-Fault)**

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in oblique slip plate margins such as
the western coast of north america one
plate moves laterally past another
resulting in strike slip faults
crust is neither created or destroyed in
this type of plate margin
the san andreas fault system is a
typical manifestation of this kind of
plate margin
when forces exceed the brittle strength
of rocks the rocks will break and shift
along faults the type of fault that is
produced depends on the type of stress
and the direction of the resulting
movement of the fault blocks
when shearing forces break rocks rocks
on opposite sides slide past each other
horizontally producing a strike slip
fault
such vaults are termed right lateral if
rocks on the opposite side have shifted
to the right with respect to an observer
or left lateral if the rocks have
shifted to the left
strike slip faulting along the margins
of tectonic plates creates transform
plate boundaries
such faulting is most commonly found
offsetting adjacent segments of
mid-ocean ridges on the sea floor
however transform boundaries may be
found within continental crust as well
the san andreas fault in california is a
well-known example
in the image above the right lateral
motion of the san andreas over time is
evidenced by the displacement of the
stream bed across the fault
the 130 meter offset was not produced in
a single earthquake but is the
cumulative result of dozens to hundreds
of quakes over many thousands of years
strike slip faults along transform
boundaries are not perfectly straight

over long distances
where bends occur in such faults as
along the san andreas fault east of los
angeles local forces can shift from
shear to compression or tension
such local shifts in forces have
produced the san gabriel mountains and
salt on trough respectively
English (auto-generated)

PLATE MARGINS

Divergent Plate Boundaries
- Mid-Atlantic Ridge
- Triple Junction: The Red Sea/East Africa

Convergent Plate Boundaries
- Oceanic/Continental: The Andes
- Continental/Continental: The Himalayas
- Oceanic/Oceanic: The Caribbean Islands

Conservative Plate Boundaries
- San Andreas Fault

Mid-plate Tectonic Activity
- Hawaiian Islands

SAN ANDREAS FAULT

The San Andreas Fault marks the junction between the North American and Pacific Plates. The fault is 1300 km long, extends to at least 25 km in depth, and has a north west-south east trend. It is classified as a right lateral (dextral) strike-slip fault.

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Although both plates are moving in a north westerly direction, the Pacific Plate is moving faster than the North American Plate, so the relative movement of the North American Plate is to the south east. The Pacific Plate is being moved north west due to sea floor spreading from the East Pacific Rise (divergent margin) in the Gulf of California. The North American Plate is being pushed west and north west due to sea floor spreading from the Mid Atlantic Ridge (divergent margin).

Movement along the fault is not smooth and continual, but sporadic and jerky. Frictional forces lock the blocks of lithosphere together for years at a time. When the frictional forces are overcome, the plates slip suddenly and shallow focus earthquakes are generated. Landscape and manmade features (eg rivers, fences and roads) are displaced across the fault as movement occurs. San Francisco has historically suffered significant earthquakes, notably in 1906 and 1989.

The average rate of movement along the San Andreas Fault is between 30mm and 50mm per year over the last 10 million years. If current rates of movement are maintained Los Angeles will be adjacent to San Francisco in approximately 20 million years.

GLOSSARY →

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G3.12

Summary

Be sure you understand and can explain the main features of plate tectonic boundaries, and how density differences in crustal material relate to processes at plate boundaries. In the next lesson, we will examine earthquakes and volcanoes, and their relationship to plate motions.

This concludes the Chapter 2 of the module Geosphere.

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